Experiment 6

AIM: Create Product Catalog Application using ASP.NET MVC Architecture. (Controllers, Models, Views, Routing)

- **Controller/ProductController.cs**

```
using System.Collections.Generic;
using System.Linq;
using System.Web.Mvc;
using Practical_6.Models;

namespace Practical_6.Controllers
{
    public class ProductController : Controller
    {
        // Temporary product storage
        private static List<Product> products = new List<Product>()
        {
            new Product { Id = 1, Name = "Laptop", Price = 45000 },
            new Product { Id = 2, Name = "Mobile", Price = 15000 },
            new Product { Id = 3, Name = "Keyboard", Price = 1000 }
        };

        // View all products
        public ActionResult Index()
        {
            return View(products);
        }

        // Open Add page
        public ActionResult Create()
        {
            return View();
        }

        // Save new product
        [HttpPost]
        [ValidateAntiForgeryToken]
        public ActionResult Create(Product product)
        {
            if (ModelState.IsValid)
            {
                product.Id = products.Count == 0
                    ? 1 : products.Max(p => p.Id) + 1;

                products.Add(product);

                return RedirectToAction("Index");
            }

            return View(product);
        }

        // Open Edit page
```

```
        public ActionResult Edit(int id)
        {
            Product product = products.FirstOrDefault(p => p.Id == id);

            if (product == null)
                return HttpNotFound();

            return View(product);
        }

        // Save edited product
        [HttpPost]
        [ValidateAntiForgeryToken]
        public ActionResult Edit(Product product)
        {
            if (ModelState.IsValid)
            {
                Product existing = products.FirstOrDefault(
                    p => p.Id == product.Id);

                if (existing == null)
                    return HttpNotFound();

                existing.Name = product.Name;
                existing.Price = product.Price;

                return RedirectToAction("Index");
            }

            return View(product);
        }

        // Open Delete confirmation page
        public ActionResult Delete(int id)
        {
            Product product = products.FirstOrDefault(p => p.Id == id);

            if (product == null)
                return HttpNotFound();

            return View(product);
        }

        // Delete product
        [HttpPost, ActionName("Delete")]
        [ValidateAntiForgeryToken]
        public ActionResult DeleteConfirmed(int id)
        {
            Product product = products.FirstOrDefault(p => p.Id == id);

            if (product != null)
                products.Remove(product);

            return RedirectToAction("Index");
        }
    }
}
```

- **Model/Product.cs**

```
namespace Practical_6.Models
{
    public class Product
    {
        public int Id { get; set; }

        public string Name { get; set; }

        public decimal Price { get; set; }
    }
}
```

- **View/Product/Create.cshtml**

```
@model Practical_6.Models.Product

<h2>Add Product</h2>

@using (Html.BeginForm())
{
    @Html.AntiForgeryToken()

    <p>Product Name</p>
    @Html.TextBoxFor(m => m.Name, new { required = "required" })

    <p>Product Price</p>
    @Html.TextBoxFor(m => m.Price,
        new { type = "number", min = "0.01", step = "0.01", required = "required" })

    <br />
    <br />
    <input type="submit" value="Add Product" />
}

<br />
@Html.ActionLink("Back to Product List", "Index")
```

- **View/Product/Delete.cshtml**

```
@model Practical_6.Models.Product

<h2>Delete Product</h2>

<p>Are you sure you want to delete this product?</p>

<p><b>ID:</b> @Model.Id</p>
<p><b>Name:</b> @Model.Name</p>
<p><b>Price:</b> ₹@Model.Price</p>

@using (Html.BeginForm())
{
    @Html.AntiForgeryToken()
```

```
    @Html.HiddenFor(m => m.Id)

    <input type="submit" value="Confirm Delete" />
}

<br />
@Html.ActionLink("Cancel", "Index")
```

- **View/Product/Edit.cshtml**

```
@model Practical_6.Models.Product

<h2>Edit Product</h2>

@using (Html.BeginForm())
{
    @Html.AntiForgeryToken()

    @Html.HiddenFor(m => m.Id)

    <p>Product Name</p>
    @Html.TextBoxFor(m => m.Name, new { required = "required" })

    <p>Product Price</p>
    @Html.TextBoxFor(m => m.Price,
        new { type = "number", min = "0.01", step = "0.01", required = "required" })

    <br />
    <br />
    <input type="submit" value="Update Product" />
}

<br />
@Html.ActionLink("Back to Product List", "Index")
```

- **View/Product/Index.cshtml**

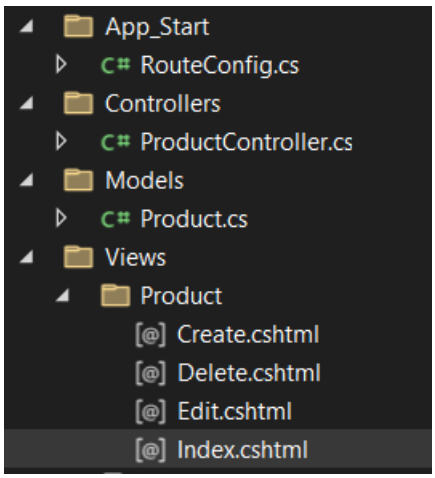
```
@model IEnumerable<Practical_6.Models.Product>

<h2>Product Catalog</h2>

<p>
    @Html.ActionLink("Add New Product", "Create")
</p>

<table border="1" cellpadding="10">
    <tr>
        <th>ID</th>
        <th>Product Name</th>
        <th>Price</th>
        <th>Actions</th>
    </tr>
```

```
    @foreach (var product in Model)
    {
        <tr>
            <td>@product.Id</td>
            <td>@product.Name</td>
            <td>₹@product.Price</td>
            <td>
                @Html.ActionLink("Edit", "Edit",
                    new { id = product.Id })
                |
                @Html.ActionLink("Delete", "Delete",
                    new { id = product.Id })
            </td>
        </tr>
    }
</table>
```



Product Catalog

Add New Product

ID	Product Name	Price	Actions
1	Laptop	₹45000	Edit | Delete
2	Mobile	₹15000	Edit | Delete
3	Keyboard	₹1000	Edit | Delete

Edit Product

Product Name

Laptop

Product Price

46000

Update Product

Back to Product List

Product Catalog

Add New Product

ID	Product Name	Price	Actions
1	Laptop	₹46000	Edit | Delete
2	Mobile	₹15000	Edit | Delete
3	Keyboard	₹1000	Edit | Delete

Delete Product

Are you sure you want to delete this product?

ID: 1

Name: Laptop

Price: ₹46000

Confirm Delete

Cancel

Product Catalog

Add New Product

ID	Product Name	Price	Actions
2	Mobile	₹15000	Edit | Delete
3	Keyboard	₹1000	Edit | Delete

Add Product

Product Name

Earburds-Pro_Tech

Product Price

1200

Add Product

Back to Product List

Product Catalog

Add New Product

ID	Product Name	Price	Actions
2	Mobile	₹15000	Edit | Delete
3	Keyboard	₹1000	Edit | Delete
4	Earburds-Pro_Tech	₹1200	Edit | Delete

GitHub Link :- https://github.com/Chirag015-create/Practical_6_DOT_NET